Yang-Jun Joo, Ph.D.

Data Scientist | Quantitative Risk Assessment

Autonomous Vehicle Safety **|** *Statistical & Probabilistic Modeling* **|** *Large-Scale Driving Data*

Orlando, FL | Open to relocation to Foster City, CA

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Professional Summary: Quantitative risk assessment data scientist with 8+ years of experience, including 3+ years of post-Ph.D. experience, developing probabilistic safety metrics, uncertainty-aware models, and automated safety risk-assessment tools for autonomous vehicle systems. Author/co-author of 15 peer-reviewed papers, with hands-on experience in Python, SQL, multi-terabyte driving data, and human behavior analysis.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, scikit-learn, PyTorch, Git, Linux, ETL pipelines.
Risk & Statistics:	Probabilistic modeling, Bayesian networks, reliability analysis, surrogate safety measures, causal inference, uncertainty quantification, conformal prediction.
Safety & Mobility:	AV risk assessment, safety-performance metrics, naturalistic driving video, connected-vehicle and trajectory data, traffic-conflict analysis, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar (Quantitative Safety Analytics) — **University of Central Florida** — Mar 2024 – Present

- Led Florida Department of Transportation safety analytics workstreams with agency and technical partners, defining and standardizing safety-performance metrics using traffic-controller, connected-vehicle, and naturalistic driving data.
- Secured KRW 73.5 million in competitive funding as principal investigator and used causal forests to quantify heterogeneous effects of vehicle automation on risk perception across 926 participants and 1,325 near-miss clips.
- Built and automated Python and SQL ETL and analytics pipelines for multi-terabyte datasets from 1,000+ intersections, standardizing risk-metric generation and reporting.
- Developed uncertainty-aware evaluation for autonomous-driving vision-language models (VLM) using conformal prediction, identifying context-specific calibration failures in safety-critical driving QA.

Postdoctoral Researcher (AV Safety Systems) — **Seoul National University** — Mar 2023 – Feb 2024

- Led research on real-time infrastructure-to-vehicle (V2I) safety information systems for the Korean National Police Agency, defining safety-critical scenarios and operational safety criteria.

Graduate Research Assistant — **Seoul National University** — Mar 2018 – Feb 2023

- Developed generalized quantitative risk assessment methods for autonomous vehicle lane changes and collision avoidance using field theory, Bayesian networks, reliability analysis, and surrogate safety measures.
- Secured and led a KRW 96.1 million Global Ph.D. Fellowship, developing and validating probabilistic AV risk-assessment methods for lane changes, collision avoidance, and driver-level crash risk.

SELECTED PUBLICATIONS

- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," *Analytic Methods in Accident Research*.
- Joo et al. (2022), "A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment," *Accident Analysis & Prevention*.
- Joo et al. (2026), "Context-Aware Conformal Prediction for VLM-Based Driving Scene QA," *ICML Workshop EMM-QA*.

EDUCATION

Ph.D., Civil & Environmental Engineering (Transportation), Seoul National University, 2023

B.S., Civil & Environmental Engineering, Seoul National University, 2018